Euler's Method and Logistic Growth

The Euler's Method formula

1 State the recursive formula used in Euler's Method to approximate solutions to $\frac{dy}{dx} = f(x, y)$.

2 Given $\frac{dy}{dx} = x + y$ with $y(0) = 1$, use Euler's Method with step size $\Delta x = 0.5$ to estimate $y(1)$.

3 Given $\frac{dy}{dx} = y$ with $y(0) = 1$, use Euler's Method with $\Delta x = 0.5$ to estimate $y(1)$, then compare to the exact solution $y = e^x$.

Euler's Method with smaller step sizes

4 Given $\frac{dy}{dx} = 2x$ with $y(0) = 0$, use Euler's Method with $\Delta x = 0.25$ to estimate $y(0.5)$, then compare to the exact solution $y = x^2$.

5 For $\frac{dy}{dx} = 2x$ with $y(0) = 0$, redo the estimate for $y(0.5)$ using $\Delta x = 0.1$ (5 steps), and compare accuracy to the $\Delta x = 0.25$ result.

Error analysis in Euler's Method

6 Explain why Euler's Method tends to underestimate y when the solution curve is concave up, using a tangent-line reasoning argument.

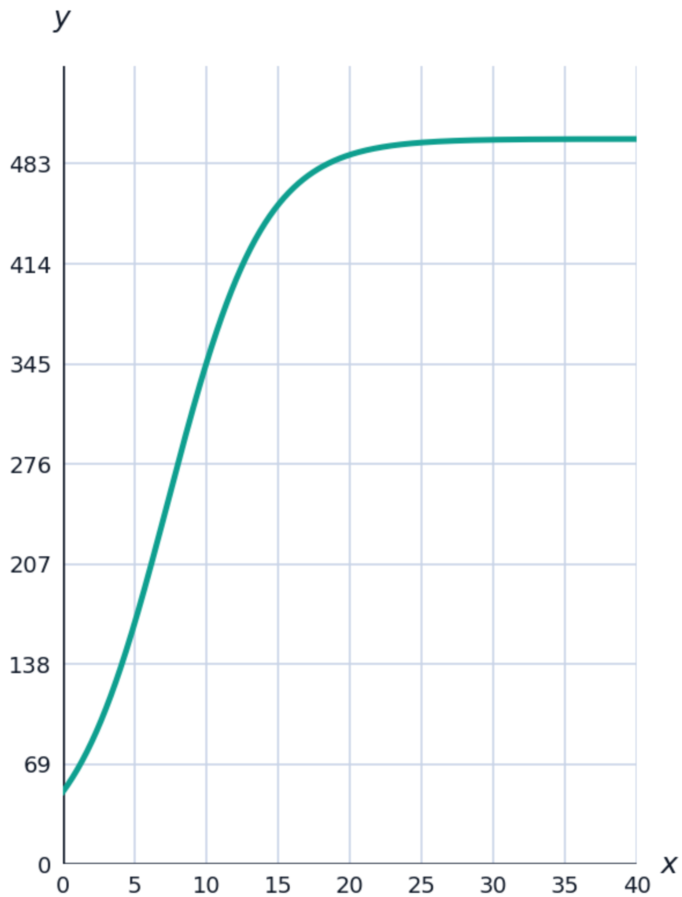
7 Explain what happens to the accuracy of Euler's Method as the step size $\Delta x \to 0$.

Logistic growth models

8 State the logistic differential equation and identify the carrying capacity L in $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$.

9 For the logistic model $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$, find the population P at which growth rate is maximized.

10 The logistic model with $L = 500$, $k = 0.3$, and $P(0) = 50$ has closed-form solution $P(t) = \frac{500}{1 + 9e^{-0.3t}}$. Sketch its shape and identify the horizontal asymptote.



11 For a logistic model with carrying capacity $L = 1000$ and $k = 0.2$, find $\frac{dP}{dt}$ when $P = 200$ and when $P = 800$, and explain which is closer to the maximum growth rate.

Interpreting logistic growth graphically

12 Explain what happens to $\frac{dP}{dt}$ as $P \to L$ in the logistic model, and what this means for the population over time.

13 Explain why the logistic model is considered more realistic than the unrestricted exponential model $\frac{dP}{dt} = kP$ for real populations.

Building an Euler's Method table

14 Given $\frac{dy}{dx} = x - y$ with $y(0) = 2$, complete an Euler's Method table with $\Delta x = 0.5$ for $x = 0, 0.5, 1, 1.5, 2$, showing each y_n.

15 Using the table above, estimate $y(2)$ and state whether the true solution is likely above or below this estimate near $x = 2$, given the slope $\frac{dy}{dx} = x - y$ becomes positive once $y < x$.

Differential equations as models of real phenomena

16 A cup of coffee cools according to Newton's Law of Cooling, $\frac{dT}{dt} = -0.15(T - 20)$, with $T(0) = 90°$C. Use Euler's Method with $\Delta t = 1$ to estimate $T(2)$.

17 Explain why Newton's Law of Cooling, $\frac{dT}{dt} = -k(T - T_{room})$, has the same mathematical structure as exponential decay shifted by a constant, and state its equilibrium (constant) solution.

Comparing Euler's Method to the exact logistic solution

18 For $\frac{dP}{dt} = 0.3P\left(1 - \frac{P}{500}\right)$ with $P(0) = 50$, use Euler's Method with $\Delta t = 2$ to estimate $P(2)$, then compare to the exact value $P(2) = \frac{500}{1 + 9e^{-0.6}} \approx 121.9$ (using the closed-form solution from earlier).

Autonomous differential equations and equilibria

19 A differential equation is called autonomous if $\frac{dy}{dx}$ depends only on y, not on x. Explain why both $\frac{dy}{dx} = ky$ and the logistic equation are autonomous, and find the equilibrium solutions of $\frac{dy}{dx} = y(y - 4)$.

20 For $\frac{dy}{dx} = y(y - 4)$, determine whether the equilibrium $y = 0$ is stable or unstable by checking the sign of $\frac{dy}{dx}$ just above and below $y = 0$.